

JAWLESS FISH

Unlike all other vertebrates, jawless fish have no biting jaws, although they do have teeth. Once abundant and diverse, few living representatives remain.

There are two living groups of jawless fish – lampreys and hagfish – and the relationship between them is still debated. Instead of jaws, lampreys have a round, sucker mouth encircled by rasping teeth, while hagfish have a slit-like mouth with teeth inside on the tongue. Circular gill openings run along the sides, seven in lampreys and a variable number (1–16) in hagfish. For body support and muscle attachment, both groups have a simple, rod-like structure called a notochord. In addition, lampreys may have a few cartilage supports around the notochord.

DIFFERING LIFECYCLES

While hagfish live permanently in the deep ocean, lampreys inhabit temperate coastal waters and fresh waters worldwide. All lampreys breed in fresh water. Coastal species are anadromous – that is, like salmon, they swim up rivers into fresh water to spawn, after which they die. The eggs hatch into worm-like, burrowing larvae, known

as ammocoetes. These live in mud and feed on detritus. After about three years the larvae of the anadromous species change into adults and swim out to sea where they feed for several years. The freshwater species live and breed in rivers and lakes. Hagfish lay eggs on the seabed.

PARASITES AND SCAVENGERS

Lampreys are notorious for feeding parasitically on larger fish when they are adult. Clinging on with its sucker mouth, a lamprey grinds its way through the skin of its victim eating either, or both, flesh and blood. They can be a nuisance to fishermen as they damage and kill fish in nets and fish farms. However, many lampreys, especially freshwater species, just eat small invertebrates. The sucker mouth is also useful when the fish are swimming upstream against the current – they use it to cling onto rocks so they can have a rest and to move pebbles when excavating a nest site in the river bed.

Hagfish live in mud burrows in the seabed, emerging at night to feed on both live and dead invertebrates. If the opportunity arises, they will feed on rotting whale or fish carcasses, from which they rasp off bits of flesh. If attacked, hagfish produce copious, gill-clogging slime to deter the predator.

PHYLUM	CHORDATA
CLASSES	PETROMYZONTI
	MYXINI
ORDERS	2
FAMILIES	4
SPECIES	128

DEBATE

ARE LAMPREYS AND HAGFISH RELATED?

Traditionally lampreys and hagfish were classified together as Cyclostomata, separate from jawed vertebrates. Detailed morphological comparisons and early molecular work allied lampreys with jawed vertebrates, leaving hagfish as a non-vertebrate chordate. Since about 2010, advanced genetic techniques (of micro-RNAs) have provided strong evidence for grouping them together once again.

LAMPREYS AND HAGFISH

The orders Petromyzontiformes and Myxiniformes are the only two groups of living jawless fish – often called agnathans. They share an eel-like shape, lack of jaws, and possession of a notochord, but also have numerous anatomical differences. Lampreys have a long larval stage spent in fresh water, while hagfish lay eggs on the seabed that hatch into miniature adults.

